

Partial Progress on Low-Dimensional DDPM Adaptivity in Wasserstein Distance

June 25, 2026

Abstract

Huang, Wei and Chen proved that the original denoising diffusion probabilistic model (DDPM) is nearly optimally adaptive to unknown intrinsic dimension when error is measured in Kullback–Leibler divergence. A follow-up open problem asks whether analogous adaptivity holds in total variation or Wasserstein distance. This note repairs a tempting KL-to-Wasserstein argument and records exactly what it proves.

The result is *partial progress, not a full solution*. Total variation follows immediately from the KL theorem by Pinsker’s inequality, although sharper direct TV rates are known. For Wasserstein distance, a KL-to- W_2 conversion is valid if the DDPM output law satisfies a quadratic transportation inequality. I prove such an inequality for exact-score DDPMs with bounded data support and expose the constant:

$$C_{\text{out}} \leq (1 + 2(T - \delta)) \exp\left(\frac{2R^2}{\sigma_\delta^2}\right), \quad \sigma_\delta^2 = 1 - e^{-2\delta},$$

where R bounds the support radius and δ is the early-stopping time. Hence any black-box bound $\text{KL}(q_\delta \| p_{\text{out}}) \leq K_N$ gives

$$W_2(q_\delta, p_{\text{out}}) \leq \sqrt{2(1 + 2(T - \delta))} \exp\left(\frac{R^2}{\sigma_\delta^2}\right) \sqrt{K_N}.$$

The estimate is dimension-free in the ambient dimension once R, δ, T are fixed, but it is not the desired Huang–Wei–Chen Wasserstein adaptivity theorem because the exponential dependence on R^2/σ_δ^2 may be enormous in their permitted parameter regime. This exponential dependence is not merely a proof artefact of the final conversion: two separated Gaussian bumps already have exponentially large T_2 constants.

Two further facts are proved. First, a dimension-free W_2 guarantee directly to the unnoised target p_{data} is impossible for the usual isotropic early-stopped comparison unless $\delta \lesssim \varepsilon^2/d$. Second, for exact scores and data supported on an affine k -plane, the DDPM recursion factorizes exactly: the orthogonal Gaussian component is transported without discretization error, so all nontrivial TV and W_2 error is confined to the k -dimensional coordinate. This gives a rigorous ambient-dimension-free reduction in the affine-subspace case.

1 Setup and status of the result

Let p_{data} be a probability measure on $\mathbb{R}^d$. The forward Ornstein–Uhlenbeck process is

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad X_0 \sim p_{\text{data}},$$

and its time- t law is denoted by q_t . Equivalently,

$$X_t = e^{-t} X_0 + \sigma_t Z, \quad \sigma_t^2 := 1 - e^{-2t}, \quad Z \sim N(0, I_d).$$

The score of q_t is $s_t = \nabla \log q_t$. A reverse DDPM sampler is run from a large time T down to a small early-stopping time $\delta > 0$, and its output law is denoted by p_{out} . The natural stopped target is q_δ . We write

$$\text{KL}(\mu\|\nu) = \int \log\left(\frac{d\mu}{d\nu}\right) d\mu, \quad \text{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|,$$

and

$$W_2^2(\mu, \nu) = \inf_{\mathcal{L}(U)=\mu, \mathcal{L}(V)=\nu} \mathbb{E}\|U - V\|_2^2.$$

The open problem motivating this note asks whether the DDPM low-dimensional adaptivity proved in KL by Huang–Wei–Chen persists when discrepancy is measured in total variation or Wasserstein distance. The following black-box hypothesis abstracts the part of their theorem used here.

Assumption 1.1 (Black-box intrinsic-dimensional KL estimate). *For the DDPM output law p_{out} , suppose that*

$$\text{KL}(q_\delta\|p_{\text{out}}) \leq K_N. \tag{1}$$

In the ideal low-dimensional regime one should think of

$$K_N \lesssim \frac{k \text{polylog}(N, A, 1/\delta)}{N} + \text{Err}_{\text{score}} + \text{Err}_{\text{init}},$$

where k is the metric-entropy intrinsic dimension and the remaining terms encode score-estimation and initialization errors. The arguments below only use the scalar bound (1).

Status. This note does not solve the full Wasserstein part of the Huang–Wei–Chen question. It proves a correct conditional KL-to- W_2 theorem with explicit constants, gives a special-case affine-subspace reduction, and identifies obstructions. The total-variation part is essentially immediate from the KL theorem and is not new at the sharp rate.

2 Total variation: immediate consequence of KL

Theorem 2.1 (Pinsker conversion). *Under [Theorem 1.1](#),*

$$\text{TV}(q_\delta, p_{\text{out}}) \leq \sqrt{\frac{K_N}{2}}.$$

Consequently, if $K_N \lesssim k \text{polylog}/N$, then $N \gtrsim k \text{polylog}/\varepsilon^2$ suffices for $\text{TV}(q_\delta, p_{\text{out}}) \leq \varepsilon$, up to the score and initialization terms contained in K_N .

Proof. Pinsker’s inequality gives $\text{TV}(\mu, \nu)^2 \leq \frac{1}{2} \text{KL}(\mu\|\nu)$. Apply it with $\mu = q_\delta$ and $\nu = p_{\text{out}}$. $\square$

Remark 2.2. This TV corollary is correct but weak. Later TV-specific analyses give $\tilde{O}(k/\varepsilon)$ iteration complexity in closely related low-dimensional settings. The point of [Theorem 2.1](#) is only that a weaker TV statement was already implicit in the KL theorem.

3 A repaired transportation route to Wasserstein bounds

A probability measure ν satisfies the quadratic transportation inequality $T_2(C)$ if

$$W_2^2(\mu, \nu) \leq 2\text{CKL}(\mu \parallel \nu) \quad \text{for every probability measure } \mu.$$

The standard Gaussian satisfies $T_2(1)$; the inequality tensorizes; and an L -Lipschitz image of a $T_2(C)$ measure satisfies $T_2(L^2C)$.

Let

$$\delta = \tau_0 < \tau_1 < \dots < \tau_N = T, \quad h_i := \tau_i - \tau_{i-1},$$

and set

$$\begin{aligned} \bar{\alpha}_i &= e^{-2\tau_i}, & \alpha_i &= \frac{\bar{\alpha}_i}{\bar{\alpha}_{i-1}} = e^{-2h_i}, & \beta_i &= 1 - \alpha_i, \\ \sigma_i^2 &:= 1 - \bar{\alpha}_i = \sigma_{\tau_i}^2, & \tilde{\beta}_i &= \beta_i \frac{1 - \bar{\alpha}_{i-1}}{1 - \bar{\alpha}_i} = \beta_i \frac{\sigma_{i-1}^2}{\sigma_i^2}. \end{aligned}$$

The score-parametrized DDPM update is

$$Y_{i-1} = F_i(Y_i) + \sqrt{\tilde{\beta}_i} \xi_i, \quad F_i(x) = \alpha_i^{-1/2} (x + \beta_i \hat{s}_{\tau_i}(x)), \quad \xi_i \sim N(0, I_d), \quad (2)$$

run for $i = N, N-1, \dots, 1$, with $Y_N \sim N(0, I_d)$.

Lemma 3.1 (Transportation constant of a Lipschitz DDPM). *Assume each map F_i in (2) is globally Lipschitz with constant L_i . Define*

$$C_N = 1, \quad C_{i-1} = L_i^2 C_i + \tilde{\beta}_i, \quad i = N, N-1, \dots, 1.$$

Then $p_{\text{out}} = \mathcal{L}(Y_0)$ satisfies $T_2(C_0)$. Moreover,

$$C_0 = \prod_{i=1}^N L_i^2 + \sum_{j=1}^N \tilde{\beta}_j \prod_{i=1}^{j-1} L_i^2, \quad (3)$$

where the empty product is one.

Proof. The vector of Gaussian inputs $(Y_N, \xi_N, \dots, \xi_1)$ is a standard Gaussian vector and satisfies $T_2(1)$. Suppose the map from the inputs $(Y_N, \xi_N, \dots, \xi_{i+1})$ to Y_i has squared Lipschitz constant at most C_i . Then the map from $(Y_N, \xi_N, \dots, \xi_i)$ to Y_{i-1} has squared Lipschitz constant at most $L_i^2 C_i + \tilde{\beta}_i$: indeed, for two input vectors, the output difference is bounded by

$$L_i \sqrt{C_i} a + \sqrt{\tilde{\beta}_i} b,$$

where a and b are the Euclidean distances in the old Gaussian inputs and in the new noise coordinate.

The squared operator norm of $(a, b) \mapsto L_i \sqrt{C_i} a + \sqrt{\tilde{\beta}_i} b$ is $L_i^2 C_i + \tilde{\beta}_i$. This proves the recursion. The Lipschitz-image property for T_2 gives the result. Expanding the recursion gives (3). $\square$

Lemma 3.2 (Exact-score derivative). *Assume $\text{supp}(p_{\text{data}}) \subseteq B(0, R)$. For the exact score,*

$$s_t(x) = -\frac{1}{\sigma_t^2} (x - e^{-t} m_t(x)), \quad m_t(x) := \mathbb{E}[X_0 \mid X_t = x],$$

and

$$\nabla s_t(x) = -\sigma_t^{-2} I_d + e^{-2t} \sigma_t^{-4} \text{Cov}(X_0 \mid X_t = x).$$

Consequently

$$\|\text{Cov}(X_0 \mid X_t = x)\|_{\text{op}} \leq R^2.$$

Proof. The formula for s_t is Tweedie's formula for the Gaussian channel $X_t = e^{-t}X_0 + \sigma_t Z$. Differentiating the posterior mean gives

$$\nabla m_t(x) = \frac{e^{-t}}{\sigma_t^2} \text{Cov}(X_0 \mid X_t = x),$$

which implies the displayed expression for ∇s_t . Finally, if $\|X_0\| \leq R$, then for every unit vector u ,

$$u^\top \text{Cov}(X_0 \mid X_t = x) u \leq \mathbb{E}[(u^\top X_0)^2 \mid X_t = x] \leq R^2.$$

□

The next proposition is the main correction to the naive proof. One should not bound all score Lipschitz constants by their worst value at δ ; doing so loses the stabilizing cancellation in the DDPM coefficient. The exact-score map has a contracting scalar part, and the remaining nonlinearity telescopes.

Proposition 3.3 (Exact-score product bound with cancellation). *Assume $\text{supp}(p_{\text{data}}) \subseteq B(0, R)$ and use exact scores in (2). Then each F_i is globally Lipschitz with a constant satisfying*

$$L_i \leq \alpha_i^{1/2} \frac{\sigma_{i-1}^2}{\sigma_i^2} \left[1 + R^2 \left(\frac{1}{\sigma_{i-1}^2} - \frac{1}{\sigma_i^2} \right) \right]. \quad (4)$$

Consequently, for every $1 \leq m \leq N$,

$$\prod_{i=1}^m L_i^2 \leq \exp \left(2R^2 \left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_m^2} \right) \right) \leq \exp \left(\frac{2R^2}{\sigma_\delta^2} \right). \quad (5)$$

Proof. By Theorem 3.2,

$$\nabla F_i(x) = \alpha_i^{-1/2} \left[\left(1 - \frac{\beta_i}{\sigma_i^2} \right) I_d + \frac{\beta_i e^{-2\tau_i}}{\sigma_i^4} \text{Cov}(X_0 \mid X_{\tau_i} = x) \right].$$

Since $h_i \leq \tau_i$, one has $\beta_i \leq \sigma_i^2$. Also

$$1 - \frac{\beta_i}{\sigma_i^2} = \frac{\sigma_i^2 - \beta_i}{\sigma_i^2} = \frac{\alpha_i \sigma_{i-1}^2}{\sigma_i^2}.$$

Thus, using $\|\text{Cov}(X_0 \mid X_{\tau_i} = x)\|_{\text{op}} \leq R^2$,

$$L_i \leq \alpha_i^{1/2} \frac{\sigma_{i-1}^2}{\sigma_i^2} + \alpha_i^{-1/2} \frac{\beta_i e^{-2\tau_i} R^2}{\sigma_i^4}.$$

The ratio of the second term to the first is

$$\frac{\alpha_i^{-1/2} \beta_i e^{-2\tau_i} R^2 \sigma_i^{-4}}{\alpha_i^{1/2} \sigma_{i-1}^2 \sigma_i^{-2}} = R^2 \frac{\alpha_i^{-1} \beta_i e^{-2\tau_i}}{\sigma_{i-1}^2 \sigma_i^2}.$$

Since $\alpha_i^{-1} \beta_i e^{-2\tau_i} = e^{-2\tau_{i-1}} - e^{-2\tau_i} = \sigma_i^2 - \sigma_{i-1}^2$, the ratio equals

$$R^2 \frac{\sigma_i^2 - \sigma_{i-1}^2}{\sigma_{i-1}^2 \sigma_i^2} = R^2 \left(\frac{1}{\sigma_{i-1}^2} - \frac{1}{\sigma_i^2} \right).$$

This proves (4). Multiplying (4) from 1 to m yields

$$\prod_{i=1}^m L_i \leq \left(\prod_{i=1}^m \alpha_i^{1/2} \frac{\sigma_{i-1}^2}{\sigma_i^2} \right) \prod_{i=1}^m \left[1 + R^2 \left(\frac{1}{\sigma_{i-1}^2} - \frac{1}{\sigma_i^2} \right) \right].$$

The first product is

$$e^{-(\tau_m - \delta)} \frac{\sigma_0^2}{\sigma_m^2} \leq 1.$$

The second product is bounded by the exponential of the telescoping sum

$$R^2 \sum_{i=1}^m \left(\frac{1}{\sigma_{i-1}^2} - \frac{1}{\sigma_i^2} \right) = R^2 \left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_m^2} \right).$$

Squaring gives (5). □

Theorem 3.4 (Certified T_2 constant for exact-score DDPM). *Assume $\text{supp}(p_{\text{data}}) \subseteq B(0, R)$ and use exact scores in the DDPM update (2). Then the DDPM output law satisfies $T_2(C_{\text{ex}})$ with*

$$C_{\text{ex}} := (1 + 2(T - \delta)) \exp\left(\frac{2R^2}{\sigma_\delta^2}\right). \quad (6)$$

Proof. By Theorems 3.1 and 3.3, every product appearing in (3) is at most $\exp(2R^2/\sigma_\delta^2)$. Moreover

$$\sum_{j=1}^N \tilde{\beta}_j = \sum_{j=1}^N \beta_j \frac{\sigma_{j-1}^2}{\sigma_j^2} \leq \sum_{j=1}^N \beta_j \leq 2 \sum_{j=1}^N h_j = 2(T - \delta),$$

because $1 - e^{-2h} \leq 2h$. Substitution in (3) proves (6). □

Corollary 3.5 (Corrected KL-to-Wasserstein conversion). *Under Theorem 1.1 and the hypotheses of Theorem 3.4,*

$$W_2(q_\delta, p_{\text{out}}) \leq \sqrt{2(1 + 2(T - \delta)) \exp\left(\frac{R^2}{\sigma_\delta^2}\right)} \sqrt{K_N}. \quad (7)$$

If, for example, $K_N \lesssim k \text{polylog}/N$, then (7) gives $W_2(q_\delta, p_{\text{out}}) \leq \varepsilon$ once

$$N \gtrsim (1 + 2(T - \delta)) \exp\left(\frac{2R^2}{\sigma_\delta^2}\right) \frac{k \text{polylog}}{\varepsilon^2},$$

up to whatever additional terms are included in the black-box quantity K_N .

Proof. The $T_2(C_{\text{ex}})$ inequality for p_{out} gives

$$W_2^2(q_\delta, p_{\text{out}}) \leq 2C_{\text{ex}} \text{KL}(q_\delta \| p_{\text{out}}).$$

Use Theorem 1.1 and take square roots. □

Remark 3.6 (Why this is not a full solution). The bound (7) is rigorous and dimension-free if R, δ, T are regarded as fixed constants. It is nevertheless not the desired general Wasserstein adaptivity theorem in the Huang–Wei–Chen setting. The broad low-dimensional regime allows the support radius to grow polynomially with the intrinsic dimension, and early stopping may be as small as an inverse polynomial. In that regime the factor $\exp(2R^2/\sigma_\delta^2)$ can be exponential in a polynomial of k . Therefore this transportation route does not prove $\tilde{O}(k \text{poly}(1/\varepsilon))$ Wasserstein complexity under only the original metric-entropy assumptions.

4 Why a transportation conversion cannot be uniformly mild

A KL estimate in the direction supplied by the black-box theorem cannot, by itself, control quadratic Wasserstein distance.

Proposition 4.1 (Small KL and TV can coexist with arbitrary W_2). *For every $0 < \eta < 1/2$ and every $M > 0$, there exist probability measures μ, ν on $\mathbb{R}$ such that*

$$\text{KL}(\mu\|\nu) \leq 2\eta, \quad \text{TV}(\mu, \nu) = \eta, \quad W_2(\mu, \nu) = M.$$

Proof. Take

$$\mu = \delta_0, \quad \nu = (1 - \eta)\delta_0 + \eta\delta_{M/\sqrt{\eta}}.$$

Then $\text{TV}(\mu, \nu) = \eta$ and

$$\text{KL}(\mu\|\nu) = \log \frac{1}{1 - \eta} \leq 2\eta.$$

The mass η of ν located at $M/\sqrt{\eta}$ must be transported to the origin, so

$$W_2^2(\mu, \nu) = \eta \left(\frac{M}{\sqrt{\eta}} \right)^2 = M^2.$$

□

Even if a transportation inequality is available, its constant cannot be uniformly mild for all low-dimensional targets. The exponential dependence on separation divided by Gaussian smoothing is already present in the target law for a two-point distribution.

Proposition 4.2 (Separated Gaussian mixtures have large T_2 constants). *Let*

$$\nu_{a,\sigma} := \frac{1}{2}N(-a, \sigma^2) + \frac{1}{2}N(a, \sigma^2)$$

on the real line, and suppose $a \geq 2\sigma$. If $\nu_{a,\sigma}$ satisfies $T_2(C)$, then

$$C \geq c a^2 \exp\left(\frac{a^2}{8\sigma^2}\right)$$

for a universal numerical constant $c > 0$.

Proof. A standard linearization of $T_2(C)$ implies the Poincare inequality

$$\text{Var}_\nu(f) \leq C \int |f'|^2 d\nu$$

for smooth Lipschitz f . It is enough to lower-bound the best Poincare constant.

Let $f(x) = -1$ for $x \leq -a/2$, $f(x) = 2x/a$ for $|x| \leq a/2$, and $f(x) = 1$ for $x \geq a/2$. Approximate this Lipschitz function by smooth functions if desired. The law $\nu_{a,\sigma}$ is symmetric, so $\mathbb{E}_\nu f = 0$. Also

$$\nu_{a,\sigma}(|X| \leq a/2) \leq \exp\left(-\frac{a^2}{8\sigma^2}\right)$$

by the elementary Gaussian tail bound. Hence $\text{Var}_\nu(f) = \mathbb{E}_\nu f^2 \geq c_0$ for a universal $c_0 > 0$. On the other hand, $|f'| = 2/a$ on $[-a/2, a/2]$ and $|f'| = 0$ elsewhere, so

$$\int |f'|^2 d\nu \leq \frac{4}{a^2} \exp\left(-\frac{a^2}{8\sigma^2}\right).$$

The Poincare constant, and hence any T_2 constant, is therefore at least $ca^2 \exp(a^2/(8\sigma^2))$. □

Remark 4.3. If $p_{\text{data}} = \frac{1}{2}(\delta_{-a} + \delta_a)$, then q_δ is a two-component Gaussian mixture with component separation $2e^{-\delta}a$ and variance σ_δ^2 . Thus even the stopped target can have an exponentially large T_2 constant when the support has separated components and δ is small. This does not rule out a direct Wasserstein analysis of DDPM, but it rules out any proof that simply combines KL with a uniformly mild T_2 constant for such targets or nearby outputs.

5 The unnoised target cannot be reached dimension-free after isotropic early stopping

The previous Wasserstein bound compares p_{out} to the stopped law q_δ . For the standard isotropic OU noising process, this distinction is essential.

Proposition 5.1 (Early-stopping obstruction). *Let $p_{\text{data}} = \delta_0$ on $\mathbb{R}^d$. Then p_{data} has intrinsic dimension zero, while*

$$q_\delta = N(0, \sigma_\delta^2 I_d)$$

and

$$W_2(p_{\text{data}}, q_\delta) = \sigma_\delta \sqrt{d} = \sqrt{d(1 - e^{-2\delta})}.$$

Consequently, if $W_2(q_\delta, p_{\text{out}}) \leq \varepsilon$, then

$$W_2(p_{\text{data}}, p_{\text{out}}) \geq \sqrt{d(1 - e^{-2\delta})} - \varepsilon.$$

To make $W_2(p_{\text{data}}, p_{\text{out}}) = O(\varepsilon)$ in this example, one must have $\delta = O(\varepsilon^2/d)$.

Proof. The only coupling between δ_0 and $N(0, \sigma_\delta^2 I_d)$ sends the Gaussian sample to zero, and the mean squared cost is $d\sigma_\delta^2$. This proves the displayed formula. The lower bound follows from the triangle inequality. Since $1 - e^{-2\delta} \asymp \delta$ for small δ , the final claim follows. $\square$

6 Affine-subspace data: exact ambient-dimension-free reduction

For data supported on an affine subspace, the DDPM recursion has an exact product structure. This does not solve the general metric-entropy problem, but it is a genuine special case in which the ambient dimension disappears from the nontrivial part of the Wasserstein error.

Let $E \subset \mathbb{R}^d$ be a k -dimensional linear subspace and let $a \in E^\perp$. Suppose

$$X_0 = (U_0, a) \in E \oplus E^\perp.$$

Set

$$\gamma_i^\perp := N(e^{-\tau_i} a, \sigma_i^2 I_{E^\perp}).$$

Proposition 6.1 (Factorization of the exact-score DDPM). *Assume p_{data} is supported on the affine subspace $a + E$ and use exact scores. Then*

$$q_{\tau_i} = q_{\tau_i}^E \otimes \gamma_i^\perp$$

and

$$s_{\tau_i}(u, v) = \left(s_{\tau_i}^E(u), -\frac{v - e^{-\tau_i} a}{\sigma_i^2} \right).$$

If the DDPM chain is initialized so that $Y_N^\perp \sim \gamma_N^\perp$, independently of the E -coordinate, then for every i the law of Y_i factorizes as

$$\mathcal{L}(Y_i) = p_i^E \otimes \gamma_i^\perp.$$

In particular,

$$\text{TV}(q_\delta, p_{\text{out}}) = \text{TV}(q_\delta^E, p_{\text{out}}^E), \quad W_2(q_\delta, p_{\text{out}}) = W_2(q_\delta^E, p_{\text{out}}^E).$$

Proof. The forward process splits orthogonally:

$$X_{\tau_i} = (e^{-\tau_i} U_0 + \sigma_i Z_E, e^{-\tau_i} a + \sigma_i Z_\perp),$$

so $q_{\tau_i} = q_{\tau_i}^E \otimes \gamma_i^\perp$. The score of a product law is the pair of the scores, and the score of $\gamma_i^\perp$ is the displayed linear function.

The orthogonal DDPM update is

$$v_{i-1} = \alpha_i^{-1/2} \left(v_i - \frac{\beta_i}{\sigma_i^2} (v_i - e^{-\tau_i} a) \right) + \sqrt{\tilde{\beta}_i} \xi_i^\perp.$$

If $v_i \sim N(e^{-\tau_i} a, \sigma_i^2 I_{E^\perp})$, then its next mean is

$$\alpha_i^{-1/2} e^{-\tau_i} a = e^{-\tau_{i-1}} a.$$

Its next variance is

$$\alpha_i^{-1} \left(1 - \frac{\beta_i}{\sigma_i^2} \right)^2 \sigma_i^2 + \tilde{\beta}_i = \alpha_i \frac{\sigma_{i-1}^4}{\sigma_i^2} + \beta_i \frac{\sigma_{i-1}^2}{\sigma_i^2} = \sigma_{i-1}^2,$$

because $1 - \beta_i/\sigma_i^2 = \alpha_i \sigma_{i-1}^2/\sigma_i^2$ and $\sigma_i^2 = 1 - \alpha_i(1 - \sigma_{i-1}^2)$. Hence the orthogonal law is exactly $\gamma_{i-1}^\perp$. The product structure follows by induction. The identities for TV and W_2 follow because both laws have the same second factor; for W_2 , the squared cost decomposes orthogonally and the common factor can be coupled identically. $\square$

Proposition 6.2 (Effect of standard Gaussian initialization in the orthogonal coordinate). *In the setting of Theorem 6.1, suppose the orthogonal initialization is an arbitrary law $\lambda_N^\perp$ rather than $\gamma_N^\perp$. Couple the orthogonal DDPM recursions with the same Gaussian noises. Then the final orthogonal discrepancy is at most*

$$A_{\delta,T} W_2(\lambda_N^\perp, \gamma_N^\perp), \quad A_{\delta,T} := e^{-(T-\delta)} \frac{\sigma_\delta^2}{\sigma_T^2}.$$

In particular, for the standard initialization $\lambda_N^\perp = N(0, I_{d-k})$,

$$W_2^2(\lambda_N^\perp, N(e^{-T} a, \sigma_T^2 I_{d-k})) = e^{-2T} \|a\|^2 + (1 - \sigma_T)^2 (d - k).$$

Proof. The difference between two orthogonal chains driven by the same noises evolves according to multiplication by

$$r_i = \alpha_i^{-1/2} \left(1 - \frac{\beta_i}{\sigma_i^2} \right) = \alpha_i^{1/2} \frac{\sigma_{i-1}^2}{\sigma_i^2}.$$

Therefore the product contraction from T to δ is

$$\prod_{i=1}^N r_i = \left(\prod_{i=1}^N \alpha_i^{1/2} \right) \frac{\sigma_0^2}{\sigma_N^2} = e^{-(T-\delta)} \frac{\sigma_\delta^2}{\sigma_T^2}.$$

Taking an optimal initial coupling gives the first claim. The displayed formula for the Gaussian initialization is the standard closed form for W_2 between Gaussians with scalar covariance matrices. $\square$

7 Conclusion

The repaired conclusions are as follows.

1. The TV part of the Huang–Wei–Chen open question is already partially implied by their KL theorem through Pinsker. This gives $\tilde{O}(k/\varepsilon^2)$ TV complexity from KL, while sharper direct TV arguments give $\tilde{O}(k/\varepsilon)$ in related settings.
2. The KL-to-Wasserstein conversion is mathematically valid only with an additional transportation inequality for the DDPM output. For exact-score DDPM with bounded support radius R , such a transportation inequality holds with the explicit constant (6). This yields the rigorous bound (7).
3. The bound (7) is not a full solution of the general Wasserstein adaptivity problem because the certified constant is exponential in R^2/σ_δ^2 . This dependence can be prohibitive under the broad parameter ranges allowed in the KL theory. Small KL alone cannot control W_2 , and separated Gaussian mixtures show that uniformly mild T_2 constants are impossible in this generality.
4. A dimension-free W_2 guarantee directly to p_{data} is impossible for standard isotropic early stopping unless δ itself scales like $1/d$.
5. For affine-subspace data and exact scores, the DDPM recursion factorizes exactly and all nontrivial error lies in the intrinsic k -dimensional coordinate. This is a genuine ambient-dimension-free reduction, but it does not yet cover arbitrary metric-entropy low-dimensional supports.

Thus the present note provides verified partial progress and a corrected theorem, not a complete resolution of the Wasserstein part of the open problem.

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